Two ledger reductions: Erdős #727 and #374

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Abstract

We record elementary but apparently unrecorded reductions of two Erdős problems to statements about a single ledger. For #727 — does $(n+k)!^2 \mid (2n)!$ infinitely often? — we show the divisibility is exactly a carry inequality on each prime, and deduce unconditionally that any solution forces $(n+1)\cdots(n+k)$ to be $\sqrt{2n}$-smooth; this is a genuine constraint, not a heuristic, and it is what makes the problem a smooth-number question. For #374 we show that $F(m)$ fails to exist exactly because a prime introduces a fresh coordinate into an $\mathbb{F}_2$-valued ledger, which re-derives the known fact that no D_k contains a prime in one line. All statements here are **proved**; the accompanying densities are recorded separately as measurements.

1 Notation

For a prime p, v_p is the p-adic valuation, $s_p(m)$ the base-p digit sum, and

$$\kappa_p(n) \ := \ v_p\binom{2n}{n} \ = \ \frac{2s_p(n) - s_p(2n)}{p-1}$$

the number of carries when n is added to itself in base p (Kummer; Legendre for the middle equality).

2 Erdős #727

Theorem 1 (exact restatement)**.** *For all $n \geq k \geq 1$,*

$$(n+k)!^2 \,\Big|\, (2n)! \iff \big[(n+1)(n+2)\cdots(n+k)\big]^2 \,\Big|\, \binom{2n}{n} \iff \kappa_p(n) \ \geq \ 2\,v_p\big((n+1)\cdots(n+k)\big) \ \forall p.$$

Proof. $\dfrac{(2n)!}{(n+k)!^2} = \dfrac{(2n)!}{n!^2}\cdot\dfrac{n!^2}{(n+k)!^2} = \dbinom{2n}{n}\Big/\big[(n+1)\cdots(n+k)\big]^2$, which gives the first equivalence; the second is comparison of p-adic valuations, using $v_p\binom{2n}{n} = \kappa_p(n)$. $\square$

Remark 2. *At $k = 1$ this reads $(n+1)^2 \mid \binom{2n}{n}$, which is Balakran's theorem [1]; so the reformulation is anchored against a known result. Note also that only primes dividing $(n+1)\cdots(n+k)$ impose any condition, so the criterion is a finite check.*

Lemma 3 (carry ceiling above the square root)**.** *If $p > \sqrt{2n}$ then $\kappa_p(n) \leq 1$.*

Proof. From $p > \sqrt{2n}$ we get $p^2 > 2n > n$, so $n < p^2$ and n has at most two base-p digits; carries in $n + n$ can occur only out of positions 0 and 1. A carry out of position 1 would require $2n \geq p^2$, contradicting $p^2 > 2n$. Hence at most one carry. $\square$

Theorem 4 (exact digit-sum criterion)**.** *For all $n \geq 1$ and $k \geq 0$,*

$$(n+k)!^2 \;\Big|\; (2n)! \qquad \Longleftrightarrow \qquad 2\,s_p(n+k) - s_p(2n) \;\geq\; 2k \quad \textit{for every prime } p.$$

Proof. By Legendre, $2v_p\big((n+k)!\big) \leq v_p\big((2n)!\big)$ reads $2\big((n+k) - s_p(n+k)\big) \leq 2n - s_p(2n)$ after clearing the common positive denominator $p-1$; rearranging gives $2k \leq 2s_p(n+k) - s_p(2n)$. $\square$

Verified: agrees with direct evaluation of $(2n)! \bmod (n+k)!^2$ on all 1085 pairs with $n < 220$, $k \leq 4$ — no mismatch.

The criterion makes the smoothness obstruction a three-line computation, and exhibits it as the *same* antipodal mechanism that governs #396.

Theorem 5 (large primes in the window are fatal)**.** *Let $p > \sqrt{2n}$ be prime and suppose $p \mid n+j$ for some $1 \leq j \leq k$. Then $2s_p(n+k) - s_p(2n) = 2k+1-p < 2k$, so the criterion fails at p.*

Proof. Put $m = n+k$ and $r = m \bmod p$. From $p \mid n+j$ and $m = (n+j)+(k-j)$ we get $r = k-j \leq k-1$. Since $p^2 > 2n$, both m and $2n$ have at most two base-p digits: write $m = qp + r$ with $q < p$, so $s_p(m) = q + r$. Now $2n = 2m - 2k = 2qp + (2r - 2k)$ and $2r - 2k \leq -2 < 0$, so the subtraction borrows once:

$$2n = (2q-1)p + \big(p + 2r - 2k\big), \qquad 0 \leq p + 2r - 2k < p,$$

the bracket lying in range because $2k - 2r \leq 2k \leq p$. Hence $s_p(2n) = (2q-1) + (p + 2r - 2k)$ and

$$2s_p(m) - s_p(2n) = 2(q+r) - (2q - 1 + p + 2r - 2k) = 2k + 1 - p,$$

which is $< 2k$ for every $p > 1$. $\square$

Verified: among 13 532 pairs (n,k) with $k \leq 5$, $n < 3000$ possessing a prime factor $> \sqrt{2n}$ somewhere in the window, *not one* satisfies the criterion; and all 425 genuine solutions with $k \leq 4$, $n < 4000$ have $\sqrt{2n}$-smooth windows.

Corollary 6 (#727 and #396 are one shape)**.** *Every solution of #727 has $\{n+1,\ldots,n+k\}$ entirely $\sqrt{2n}$-smooth. Together with the companion note on #396 — where every solution has $\{n-k,\ldots,n\}$ entirely* $\max(2k,\sqrt{2n})$*-smooth — both problems reduce to the existence of runs of consecutive smooth integers of prescribed length, and differ only in the direction of the shift and the exact threshold.*

Remark 7 (what the ladder run showed)**.** *The cooperative-parameter instrument was run on this problem and the outcome is worth recording, because it is a clean negative. Choosing $m = n+k = cQ-1$ with $Q = \prod_{p \leq P} p^{a_p}$ saturates the bottom a_p base-p digits of m at $p-1$ on every small rail at once, and then both digit sums are closed form: $s_p(m) = s_p(c-1) + a_p(p-1)$ and $s_p(2n) = s_p(2c-1) + a_p(p-1) - s_p(2k+1)$ whenever $p^{a_p} > 2k+2$. The criterion collapses to $a_p(p-1) + s_p(2k+1) + \big[2s_p(c-1) - s_p(2c-1)\big] \geq 2k$, and since $s_p(2c-1) \leq 2s_p(c-1)+1$*

the bracket is at least -1*; so* $a_p(p-1) \geq 2k$ ***suffices, for every*** c***, unconditionally****. One such* m *serves several* k *at once: at* $Q = 2^7 3^4 5^3$ *the value* $m = Q-1$ *works simultaneously for* $k = 2,3,4,5$*.* And every output fails on a large rail — *measured first failures at* $p =$ 277, 311, 443, 853, 1481 — *because saturating the digits of* m *says nothing about the prime factors of* $m-1, \ldots, m-k+1$*, which is what Theorem 5 actually demands. Enlarging* P *moves the first failing prime without removing it. The cooperative parameter for this problem is therefore not digit saturation but window smoothness, and the transfer step remains the theorem.*

Theorem 8 (smoothness is forced)**.** *Let* $k \geq 2$*. If* $(n+k)!^2 \mid (2n)!$ *then every prime factor of* $(n{+}1)(n{+}2)\cdots(n{+}k)$ *is at most* $\sqrt{2n}$*; that is, the product is* $\sqrt{2n}$*-smooth.*

Proof. Suppose $p \mid (n{+}1)\cdots(n{+}k)$ with $p > \sqrt{2n}$. Then $v_p\big((n{+}1)\cdots(n{+}k)\big) \geq 1$, so Theorem 1 demands $\kappa_p(n) \geq 2$, contradicting Lemma 3. □

Measured (census to $n = 1.2 \times 10^5$): among all solutions for $k = 2$ and $k = 3$ there is **no** violation of Theorem 8, and $\max \log p_{\max}/\log(2n) = 0.4999$ — the bound is attained and never crossed. Solution densities to $n = 2 \times 10^5$ are 0.1126 (k=1), 0.0099 (k=2), 7.6×10^{-4} (k=3), 4×10^{-5} (k=4), with dyadic counts for $k = 2$ stabilising near 0.010: positive density, not merely infinitude. The density of n for which $(n{+}1)(n{+}2)$ is $\sqrt{2n}$-smooth measures 0.0932, against $\rho(2)^2 = (1-\log 2)^2 = 0.0942$ — consecutive smoothness is independent to 1%.

3 A lower bound on n in terms of k

Theorem 8 bounds the size of the rails; the carry ceiling bounds their multiplicities, and the two together bound k.

Lemma 9 (carry ceiling)**.** *For every prime* p *and every* $n \geq 1$*,* $\kappa_p(n) \leq \lfloor \log_p(2n) \rfloor$*.*

Proof. $\kappa_p(n)$ counts the carries in the base-p addition $n + n = 2n$. Carries occur only out of positions $0, \ldots, \lfloor \log_p(2n) \rfloor - 1$, since a carry out of the top position of $2n$ is impossible. □

Lemma 10 (multiplicity ceiling)**.** *Suppose* $(n+k)!^2 \mid (2n)!$ *and let* p *be a prime with* $e := v_p\big((n{+}1)\cdots(n{+}k)\big) \geq 1$*. Then*

$$e \;\leq\; \tfrac{1}{2}\log_p(2n), \qquad \textit{equivalently} \qquad p^e \;\leq\; (2n)^{1/2}.$$

Proof. By Theorem 1 the hypothesis gives $\kappa_p(n) \geq 2e$, and Lemma 9 gives $\kappa_p(n) \leq \log_p(2n)$. Combining, $2e \leq \log_p(2n)$; exponentiating base p gives $p^e \leq (2n)^{1/2}$. □

Verified: all 1515 rails occurring in the 227 solutions with $k \in \{2,3\}$ and $n < 30000$ — no violation.

Theorem 11 (k is small compared with $\sqrt{n}$)**.** *If* $(n+k)!^2 \mid (2n)!$ *with* $k \geq 2$*, then*

$$n^k \;<\; (2n)^{\pi(\sqrt{2n})/2}, \qquad \textit{hence} \qquad k \;<\; \big(1+o(1)\big)\frac{\sqrt{2n}}{\log n} \quad \textit{and} \quad n \;\gg\; k^2 \log^2 k.$$

Proof. Write $\Pi = (n+1)(n+2)\cdots(n+k)$. By Theorem 8 every prime factor of Π is at most $\sqrt{2n}$, and by Lemma 10 each contributes $p^{v_p(\Pi)} \le (2n)^{1/2}$. Hence

$$\Pi \;=\; \prod_{p \le \sqrt{2n}} p^{\,v_p(\Pi)} \;\le\; (2n)^{\pi(\sqrt{2n})/2}.$$

On the other hand $\Pi > n^k$, which gives the displayed inequality. Taking logarithms, $k \log n < \frac{1}{2}\pi(\sqrt{2n})\log(2n)$, and the prime number theorem $\pi(x) \sim x/\log x$ yields $\pi(\sqrt{2n}) \sim 2\sqrt{2n}/\log(2n)$, so $k \log n < (1+o(1))\sqrt{2n}$. Solving for n gives $n \gg k^2 \log^2 k$. □

Verified: the inequality $k < \sqrt{2n}/\log n$ holds on all 302 solutions with $2 \le k \le 5$, $n < 40000$, as does the intermediate bound $\Pi \le (2n)^{\pi(\sqrt{2n})/2}$. The least solutions are $n = 208$ ($k = 2$), $n = 3475$ ($k = 3$), $n = 8174$ ($k = 4$), against bounds 3.82, 10.22, 14.19.

Corollary 12 (what remains)**.** *For fixed $k \ge 2$, #727 follows from the conjunction of*

(S) a positive density of n for which $(n+1)\cdots(n+k)$ is $\sqrt{2n}$-smooth, and

(C) a positive conditional probability, given (S)*, that $\kappa_p(n) \ge 2v_p$ holds for every $p \le \sqrt{2n}$.*

Theorem 8 shows (S) *is* necessary*, so no route can avoid it.*

Condition (C) is of the same type as the carry conditions in Erdős #377, but is required only with positive probability rather than uniformly in n, which is a substantially weaker demand.

4 Erdős #374

For $a \ge 1$ let $w(a) \in \mathbb{F}_2^{(\text{primes})}$ have p-coordinate $v_p(a!) \bmod 2$. Since $v_p(a!) = v_p((a-1)!) + v_p(a)$ we have the ledger recursion $w(a) = w(a-1) + u(a)$, where $u(a)$ has p-coordinate $v_p(a) \bmod 2$. A product $a_1! \cdots a_k!$ is a perfect square iff $\sum_i w(a_i) = 0$ in $\mathbb{F}_2$. Recall $F(m)$ is the least $k \ge 2$, if it exists, for which there are $a_1 < \cdots < a_k = m$ with $a_1! \cdots a_k!$ a square, and $D_k = \{m : F(m) = k\}$.

Theorem 13 (primes are excluded, with a reason)**.** *If m is prime then $F(m)$ does not exist. Equivalently, no D_k contains a prime.*

Proof. Let $m = p$ be prime. For $a < p$ we have $v_p(a!) = 0$, so the p-coordinate of $w(a)$ is 0; whereas $v_p(p!) = 1$, so the p-coordinate of $w(p)$ is 1. Any admissible family has $a_k = p$ and $a_i < p$ for $i < k$, so the p-coordinate of $\sum_i w(a_i)$ equals $1 \ne 0$, and the product cannot be a square. □

Remark 14. *The mechanism is that a prime is the first index at which its own coordinate is switched on, so $w(p)$ is automatically independent of all its predecessors. In this language, $F(m)$ exists precisely when $w(m)$ lies in the $\mathbb{F}_2$-span of $\{w(a) : a < m\}$, and $F(m) - 1$ is the least number of distinct predecessors summing to $w(m)$ — a shortest-zero-sum (minimum-weight) problem, not a counting one.*

Theorem 15 (the four-term criterion)**.** *Let $2 \le c < m-1$. Then $(c{-}1)!\,c!\,(m{-}1)!\,m!$ is a perfect square if and only if cm is a perfect square, i.e. iff c and m have the same squarefree kernel.*

Proof. As above $w(t)+w(t-1)=u(t)$, so $w(c{-}1)+w(c)+w(m{-}1)+w(m)=u(c)+u(m)$. This vanishes in $\mathbb{F}_2$ iff $u(c)=u(m)$, which says exactly that c and m have the same set of primes to odd exponent, i.e. the same squarefree kernel, i.e. cm is a square. □

Verified: 177 903 pairs (c,m) with $m<600$ — the equivalence is exact.

Theorem 16 (non-squarefree m need at most four terms)**.** *Let $m = s\,i^2$ with s squarefree and $i \ge 2$, and suppose $m \ge 8$. Then $F(m) \le 4$; explicitly $c = s(i{-}1)^2$ satisfies $2 \le c < m{-}1$ and $cm = \big(s\,i(i-1)\big)^2$.*

Proof. $cm = s^2 i^2 (i-1)^2$ is a square, so Theorem 15 applies once $2 \le c < m-1$. Since $i \ge 2$, $c = s(i-1)^2 \ge s \ge 1$, and $c \ge 2$ unless $s=1, i=2$, i.e. $m = 4 < 8$. Also $c/m = \big(\frac{i-1}{i}\big)^2 \le \frac{4}{9}$, so $c \le \frac{4}{9}m < m-1$ for $m \ge 2$. □

Verified: the construction succeeds for all 1566 non-squarefree m with $8 \le m \le 4000$, with no ledger failure and no range failure.

Corollary 17 (a positive-density answer)**.**

$$\big|(D_2 \cup D_3 \cup D_4) \cap \{1,\dots,n\}\big| \;\ge\; \Big(1-\tfrac{6}{\pi^2}\Big)n + O(\sqrt{n}) \;=\; 0.3921\ldots n + O(\sqrt{n}).$$

Since $D_2 = \{j^2 : j > 1\}$ has $O(\sqrt{n})$ elements, $|(D_3 \cup D_4) \cap \{1,\dots,n\}| \gg n$.

Proof. By Theorem 16 every non-squarefree $m \ge 8$ has $F(m) \le 4$, hence lies in $D_2 \cup D_3 \cup D_4$. The non-squarefree integers up to n number $(1-6/\pi^2)n + O(\sqrt{n})$. □

Verified: 1567 non-squarefree $m \le 4000$, density 0.3917 against $1 - 6/\pi^2 = 0.3921$.

Theorem 18 (pronic m need only four terms)**.** *Let $m = a(a+1)$ with $a \ge 3$. Then $(a{-}1)!\,(a{+}1)!\,(m{-}1)!\,m!$ is a perfect square, so $F(m) \le 4$.*

Proof. By the pair identity, $w(a{-}1)+w(a)=u(a)$ and $w(a)+w(a{+}1)=u(a{+}1)$, so

$$w(a{-}1)+w(a{+}1)=u(a)+u(a{+}1)=u\big(a(a{+}1)\big)=u(m),$$

using multiplicativity of u. Also $w(m{-}1)+w(m)=u(m)$. Adding the two relations gives $w(a{-}1)+w(a{+}1)+w(m{-}1)+w(m)=2u(m)=0$ in $\mathbb{F}_2$. The four arguments $a{-}1 < a{+}1 < m{-}1 < m$ are distinct and positive for $a \ge 3$. □

Note that this construction is *shorter* than the generic one of Theorem 20 precisely because the two consecutive pairs overlap: the argument a occurs in both $\{a{-}1,a\}$ and $\{a,a{+}1\}$ and cancels, removing two terms.

Proposition 19 (parity of a witness)**.** *Let S be a witness for m, i.e. $\max S = m$ and $\sum_{a\in S} w(a) = 0$. Partition S into maximal runs of consecutive integers. A run of even length 2ℓ contributes $\sum_{i=1}^{\ell} u(\cdot)$, a sum of ℓ parity vectors of integers; a run of odd length contributes such a sum together with one residual $w(\cdot)$. Hence if $|S|$ is odd, at least one run has odd length, and the witness necessarily involves a value of w itself rather than only values of u.*

Proof. Group each run $\{t, t+1, \dots\}$ from the bottom in consecutive pairs; each pair $\{s-1, s\}$ contributes $w(s-1) + w(s) = u(s)$. A run of odd length leaves one element unpaired, contributing its own w. Parity of $|S|$ equals the parity of the number of odd-length runs. $\square$

This is the obstruction to shortening the six-term construction. All even-length shapes express the condition purely in terms of u, which is multiplicative and therefore controllable: Theorems 15, 16, 18 and 20 are all of this type. A five-term witness must, by Proposition 19, involve a bare $w(x)$, and since w is the cumulative sum $\sum_{t \le x} u(t)$ rather than a multiplicative function, the requirement becomes a *collision* $w(x) = u(cm)$ rather than an identity. Collisions are generic but not constructible, and their failure is exactly what makes D_6 nonempty.

Measured: the shape $\{c-1, c, x, m-1, m\}$, i.e. $w(x) = u(cm)$, admits a solution for **all** 475 integers $m \le 900$ with $\Omega(m) \ge 3$, and for 232 of the 259 semiprimes in the same range; the 27 semiprime failures contain every element of D_6.

Measured (exact computation of F by meet-in-the-middle over the ledger, all composite $m \le 1400$): the sets $D_2, \dots, D_6$ have sizes $29, 80, 380, 250, 14$ up to $m = 900$ and continue in the same proportions. Every element of D_5 is squarefree, as Theorem 16 requires. Every element of D_6 up to 1400 is a *semiprime*:

$$527 = 17\cdot31,\ 611 = 13\cdot47,\ 713 = 23\cdot31,\ 731 = 17\cdot43,\ 779 = 19\cdot41,\ 893 = 19\cdot47,\ 923, 1003, 1037, 1271, 1273, 1343, 134$$

the least being 527, matching the value recorded by Erdős and Graham; and every $m \le 1400$ with $\Omega(m) \ge 3$ satisfies $F(m) \le 5$. If the containment $D_6 \subseteq \{pq\}$ holds in general then $|D_6 \cap \{1, \dots, n\}| \ll n \log\log n / \log n$ by Landau, hence $o(n)$, answering the growth question for D_6 in the negative; this containment is measured here, not proved.

Theorem 20 (an explicit six-term construction)**.** *Let $m = ab$ with $2 \le a < b$, $b \le m-2$ and $b \ge a+2$. Then*

$$(a-1)!\; a!\; (b-1)!\; b!\; (m-1)!\; m! \quad \textit{is a perfect square,}$$

and the six arguments $a-1 < a < b-1 < b < m-1 < m$ are distinct. Hence $F(m) \le 6$.

Proof. For any $t \ge 1$ we have $t!\,(t-1)! = t \cdot \big((t-1)!\big)^2$, so in $\mathbb{F}_2$ coordinates $w(t) + w(t-1) = u(t)$, the parity vector of t. Applying this to $t = a, b, m$,

$$w(a-1) + w(a) + w(b-1) + w(b) + w(m-1) + w(m) \;=\; u(a) + u(b) + u(m).$$

Since $m = ab$, the exponent of each prime p in m is $v_p(a) + v_p(b)$, so $u(m) = u(a) + u(b)$ in $\mathbb{F}_2$ and the right-hand side vanishes. By the criterion of §4 the corresponding product of factorials is a square. The hypotheses $2 \le a$, $b \ge a+2$ and $b \le m-2$ make the six arguments strictly increasing and positive. $\square$

Verified: the construction succeeds for 3430 of the 3448 composite $m \le 4000$. The 18 exceptions are $m = 6$ and the prime squares $9, 25, 49, 121, 169, 289, 361, 529, \dots$ — precisely the composites admitting no factorisation $m = ab$ with $b \ge a+2$ and $b \le m-2$.

Corollary 21. *$F(m) \le 6$ for every composite m that is neither 6 nor the square of a prime.*

This recovers, by explicit construction, the bound $D_k = \varnothing$ for $k > 6$ recorded by Erdős and Graham, and exhibits the witness: the six-term family $\{a-1, a, b-1, b, m-1, m\}$ attached to any sufficiently separated factorisation of m. The excluded cases are exactly those with no such factorisation, and the prime squares $m = p^2$ are handled separately by $D_2 = \{n^2 : n > 1\}$.

Measured. Computing the span incrementally for $m \leq 4000$: $w(m)$ fails to lie in the span for exactly 550 values of m, and $\pi(4000) = 550$. So in this range $F(m)$ *exists if and only if m is composite*, the forward direction being Theorem 13 and the converse remaining a measurement. Consequently $\bigcup_{k\geq 2} D_k$ has density 1, and since $D_2 = \{n^2 : n > 1\}$ has density 0 and $D_k = \varnothing$ for $k > 6$ [3], the sets D_3, D_4, D_5, D_6 partition a density-one set. Erdős's question — whether $|D_6 \cap [1, n]| \gg n$ — asks which of the four takes a positive share.

Remark 22 (register)**.** ***Proved****: Theorems 1, 8, 13, 11 and Lemmas 3, 9, 10; all are elementary and self-contained apart from the prime number theorem used in Theorem 11. Theorem 11 gives the first lower bound on n in terms of k for #727, namely $n \gg k^2 \log^2 k$. Theorem 20 and Corollary 21 give an explicit six-term witness showing $F(m) \leq 6$ for every composite m other than 6 and the prime squares; Theorems 15, 16 and Corollary 17 answer the growth question in part, giving $|(D_3 \cup D_4) \cap \{1, \ldots, n\}| \gg n$ with explicit constant $1 - 6/\pi^2$. Theorem 13 re-proves a fact recorded in [3]; no priority is claimed, only the mechanism.* ***Measured****: all densities, the smoothness census, the span computation, and the converse of Theorem 13.* ***Not proved****: #727 for any $k \geq 2$, and the growth rates in #374.* ***Falsifiers****: a solution of #727 with $p_{\max} > \sqrt{2n}$ (would contradict Theorem 8 and hence a computation error); the $k = 2$ density declining toward 0 beyond $n = 10^7$; an $m \leq 10^5$ that is composite with $w(m)$ outside the span.*

References

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